



High-Level System Architecture

Your system is a **Centralized Cloud Platform** where all roles interact with one shared database.

- **The Backend (The Engine):** Houses the logic for routing, inventory, and GenAI.
- **The Frontend (The Dashboards):** Six different "Views" tailored to each role.
- **The Database (The Heart):** Stores all live data (GPS, Stock, Labor, Payments).



The Sequence of Roles (Step-by-Step Flow)

This sequence follows the lifecycle of a **Personal Move (House Shift)**, which is your most complex scenario.

Step 1: The Requester (Individual/Vendor)

- **Action:** The user logs in and books a "Shift."
- **Data Input:** Selects destination, number of **Laborers**, and **Packing Utensils**.
- **System Result:** An **ORDER_ID** is created with a "Pending" status.

Step 2: The Logistics Manager (Owner)

- **Action:** System automatically notifies the Manager of a new high-value booking.
- **Data Input:** Oversees the transaction and ensures a warehouse is assigned to handle the request.
- **System Result:** The order is routed to the nearest **Warehouse Manager**.

Step 3: The Warehouse Manager

- **Action:** Receives the order.
- **Data Input:** 1. Assigns specific **Laborers** from the roster.
2. Prepares the **Utensil Kit** (boxes, tape).
3. Packs any inventoried items.
- **System Result:** Updates status to "Ready for Dispatch."

Step 4: The Dispatcher

- **Action:** Matches the "Ready" order to a vehicle.
- **Data Input:** 1. Selects a truck with enough seats for the **Laborers**.
2. Calculates the optimized route and "Move Time."
- **System Result:** Pushes the digital manifest to the **Driver**.

Step 5: The Driver

- **Action:** Arrives at the Warehouse, picks up the **Laborers** and **Utensils**.
- **Data Input:** 1. Clocks in the laborers.
2. Navigates to the Individual's home.
3. Performs the move/packing.
4. Uploads **Proof of Delivery (PoD)** photo.
- **System Result:** Updates status to "Completed."

Step 6: Customer Support (The Bot)

- **Action:** Monitors the entire journey.
- **Interaction:** 1. Sends the "Driver is 5 mins away" alert.
2. Asks the customer for a rating after completion.
3. Answers any "Where is my crew?" questions.
- **System Result:** Closes the feedback loop and updates the **Driver's Rating**.



Role Interaction Matrix

Sequence	Role	Primary Tool	Output to Next Role
Start	Individual	Booking UI	Order Data
Control	Manager	Admin Dashboard	Approved Workflow
Resource	Warehouse	Inventory/Staff Map	Packed Box + Crew
Logic	Dispatcher	Route Map	Assigned Route
Execution	Driver	Mobile App	Proof of Delivery
Support	AI Bot	Chat Interface	Customer Rating